

PROTECT-DB: Protecting Data using Replicated State Machines: Efficient Corruption Detection & Recovery

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Abstract

Data is critical for the operation of any organization and needs to be protected, especially against attacks that compromise the state of the database. In this paper, we explore an approach based on Byzantine-fault tolerant replicated state machines, built on top of a deterministic extension of PostgreSQL. Each replica deterministically executes transactions recorded in a shared log/blockchain. Our focus is on creating a practical system that is designed for efficient and quick detection of corruption, as well as quick repair concurrent with execution of transactions. We also present a performance study showing the efficiency and practicality of our approach. We believe our work lays the foundations for the practical use of the BFT replicated state machine approach in the context of databases.

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1 Introduction

Data is critical for any organization, and there have been numerous incidents in recent years where attackers have broken-in and damaged data. Most databases support read/write access control. However, such access control does not protect against users who have administrator privileges on the database; in fact, most applications connect to databases via logins that can update any of the application data. A malicious user with administrative privileges could compromise the database contents. This could happen due to an intruder getting administrator credentials, or due to an insider attack. In this paper, we address the problem of detection and repair of such corruption of data.

One approach to handling the problem is to use a set of database replicas, with the security of each node configured independently with its own administration. It is harder for the attacker to break into the security of multiple nodes, thus providing protection from corruption as long as only a minority of nodes is compromised. If the replicas are under the same administration, a compromised user account will result in the compromise of all replicas. It may be noted that replication is used to provide redundancy in the event

of hardware failure, which can also be used for protection from attacks.

However, in typical implementations of database replication, updates happen at a single node and are then replicated to the other nodes. If the source of the update is compromised, data in all the replicas will also be compromised. Executing the updates independently on the database replicas is not an option since the transaction ordering may differ on different nodes, and further, some transactions may commit on one node but not on another.

The approach we follow is based on Byzantine-fault tolerant replicated state machines, where transactions are recorded in a shared log using consensus mechanisms, and are then executed on all replicas.

There are three major challenges in designing such a system. First, all nodes need to agree on the order of transactions submitted to the system. Second, all nodes must be kept synchronized while executing the same sequence of transactions. Third, we need to efficiently identify and recover from corrupted data/compromised nodes.

The first challenge, of ordering input requests, is addressed by using a shared log, through distributed consensus protocol (such as Byzantine consensus protocols), or permissioned block chains (with each block containing only 1 transaction, to minimize latency) can be used instead. The log may alternatively be created by an external system, for example, a matching engine in a stock market, with the transactions being recorded persistently at a share depository. All database replicas fetch transactions from this shared log and execute them independently.

The second challenge, of keeping databases synchronized, arises from the concurrent execution of transactions, which can result in different serialization orders in different replicas. Serial execution avoids this problem, but is too inefficient to be used in practice. Deterministic database models [1], [28], [26] allow some amount of concurrent execution for efficiency, but guarantee that the serialization order will be the same, and hence the result as well as the final state will be the same across all replicas.

The third and final challenge, which is the focus of this paper, is how to identify compromised nodes/corrupted data, and recover from corruption efficiently, even on very large databases.

In this paper, we describe an architecture for a system designed to deal with Byzantine data corruption, with a focus on simple, efficient, and practical schemes for corruption detection and recovery.

Our specific contributions are as follows.

- (1) We present (in Section 3) our system model, which provides a practical architecture for building database applications, by using a BFT replicated state machine approach on top of existing databases. The model, also used in some other

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systems such as [33], is based on transaction code being executed at servers, while clients submit transaction requests that are basically API calls with parameters, digitally signed by the client.

We also present (in Section 4) a deterministic execution algorithm, which is based on Aria [28].

- (2) We present our corruption detection and recovery techniques in Section 5.
 - (a) We describe two threat models (Section 5.1), one where database data is corrupted but system code is safe, and another where system code also could be compromised.
 - (b) We discuss (in Section 5.2) two schemes to detect corruption of data. The first scheme (the passive scheme) checks the results of every query to detect any divergence, which would happen if some nodes get compromised and the data in the nodes get corrupted. If the majority of nodes are up and agree on the result of a transaction, that is accepted as the outcome. A node is identified as compromised if it gives a different result than the majority.
The second scheme (the active scheme) checks for divergence of database state across replicas, even if no query accesses the parts that have diverged. We compare database states on all nodes using a Merkle tree on each relation. To reduce Merkle root contention, we partition the records across multiple Merkle trees per relation.
 - (c) We propose (in Section 5.3) an efficient and practical scheme for recovering from corruption, which can execute even as the system continues to process other transactions. This scheme depends on getting database snapshots, a feature that is supported by many databases such as PostgreSQL.
- (3) Replicated state machines depend critically on transactions being deterministic; yet bugs can cause non-deterministic behavior, which in turn can cause database state divergence. We propose (Section 7) a simple scheme for quick detection and recovery from execution of non-deterministic transaction, which can cause serious problem if it occurs and is not detected and fixed immediately.
- (4) We have implemented all our techniques. Section 9 describes details of implementation including consensus and support for clients to submit transactions and get results back from the replicas, and support for recovery. We use Kafka for fault-tolerant asynchronous messaging. We also made small extensions to PostgreSQL to support deterministic execution based on the rw-conflict detection mechanism of Aria [28], implemented in PostgreSQL by AriaBC (github.com/zllai/AriaBC), and to implement Merkle trees.
- (5) We present a performance study in Section 10 that demonstrates that our implementation is very efficient, and can support a throughput of over 2500 transactions per second from the YCSB benchmark even on low-end hardware. Our study shows that the overheads of deterministic execution and Merkle trees are quite low compared to the baseline performance of PostgreSQL. Distributed consensus

and result-persistence do impose overheads, and optimizing their performance to further improve throughput is an area of future work.

Our approach provides a practical and effective way to detect and recover from unauthorised updates to the database. We believe it can have significant real-world impact. Preventing unauthorised leakage of information is a separate problem and not the focus of this paper.

2 Related Work

We discuss related work on Byzantine-fault tolerant replicated state machines in Section 2.2, related work on deterministic execution, in Section 2.4. and other related work in Section 2.5.

2.1 Blockchain

Blockchain technology is based on decentralization and scalability, and achieves consistency by building an immutable log through consensus [9], [59]. Blockchains support simple operations like account or transaction lookup but are not suited for more complex queries. Research [53] has shown some improvement in query performance by creating b-tree index but it has not been adopted in mainstream deployment. Delta [16] proposes a mechanism to synchronize the state of a distributed ledger to the database (NoSQL and relational), so that complex queries can be run efficiently. It discusses analogy of blockchain smart contracts to database stored procedures, which are invoked by transactions.

EngraveChain [43] describes a tamper-proof distributed log system, which can also provide privacy by encrypting data before writing to blockchain. They also propose to store large files off-the-chain with only file-hash on the chain. However, they rely on a change in system behavior (like slower performance) to detect break-ins, which is unlikely to detect subtle tampering. Also, they do not provide detail recovery procedure to synchronize state from healthy replica as we do.

Merkle trees are extensively used in commercial blockchain deployments [15], [3] to compactly represent the system state. They use (a) hashing functions which generate fixed length output (used as key in KV-stores like LevelDB or RocksDB etc) and (b) serialization schemes, which flatten nested arrays of data, e.g. space-efficient RLP (Recursive Length Prefix) encoding. Bitcoin and Algorand [47] use Merkle trees with SHA algorithm, while Ethereum uses a variant namely, Merkle Patricia tree with Keccak-256 algorithm. In a blockchain like Ethereum with millions of leaf nodes, a branch node with all child pointers becomes too large, hence its content is stored as value, while its corresponding hash is used as key. Thus, when searching for a leaf node starting with root node, multiple database lookups are needed traversing down the tree. Since we configure merkle tree with fixed height and branching, it can reside in memory thus making traversal faster.

Multiple methods [11], [41], [7], [27] have been proposed to execute intra-block transactions concurrently in Ethereum. To arrive at a serializable concurrent schedule, a miner may either delay or rollback transactions having data conflicts, as determined by performing static-analysis of transactions etc. Multiple miners may determine distinct schedules and only most optimal proposal (with most parallel validation) will be accepted by consensus as the next

block, whereas miners with suboptimal proposals will not be compensated. Also, the winning miner can order transactions in some arbitrary serial schedule, not necessarily honoring the input order. In contrast, in our proposed approach the replicas execute transactions deterministically based on data conflicts, but strictly as per their order in the input log as defined by the consensus leader. In our approach the merkle tree is partitioned, reducing the contention among concurrent transactions trying to update the root state, while slightly increasing the cost of state comparison across replicas. Since the data in Merkle Patricia tree is encoded and cryptographically chained, recovering a corrupted node with tampered data is done through full copy.

A Prolly Tree (Probabilistic B-tree) [14] is a content-addressed B-tree that provides fast read/write performance. Data is sorted by key and divided into chunks using a deterministic, content-defined chunking algorithm. Each chunk receives a content hash, which is used to identify the node and construct higher levels of the tree. They offer an important property of history independence: the same set of key-value pairs produces the same tree regardless of insertion, update, or deletion order. This determinism makes equivalent data structures have identical hashes, enabling efficient comparison and storage sharing. They support search and merkle hashing. Since we use PostgreSQL, our merkle tree does not need to support searching.

Apache Cassandra [17] uses Merkle trees for anti-entropy repair, allowing replicas to efficiently identify and reconcile inconsistent data. Each replica constructs a binary Merkle tree in which leaves contain hashes of row values and internal nodes contain hashes of their child nodes. Replicas exchange and compare these trees; when a hash differs, Cassandra recursively examines the corresponding subtree to locate the inconsistent data range and streams the required data to repair the replica. Cassandra uses a compact Merkle tree with a depth of 15 (up to 32K leaves), trading finer-grained difference detection for lower memory usage and faster tree exchange. Repair is performed using nodetool repair, with incremental repair able to build trees only for unrepaired SSTables, reducing repeated work. Cassandra’s Merkle trees are primarily temporary structures for detecting replica divergence, rather than history-independent deterministic tree construction. In contrast, our databases and thus merkle trees can be much larger, hence we use dynamic trees instead of fixed size tree. While they send all tree data to compare in a single request, we suggest a multi-level approach that balances disk-fetch time with the data transfer time.

2.2 Byzantine Fault Tolerance

Byzantine-fault tolerant replicated state machines are a well-known approach for keeping data safe from attacks. See [8], [13] for a survey of the area from a systems perspective.

In a replicated state machine, a common input log of transactions is read by each of multiple replicas, and processed deterministically in the input log order. Depending on the application needs, the log could be generated by a crash tolerant consensus protocol (like Raft), or by a Byzantine-fault tolerant consensus protocol, such as those used in (permissioned) blockchains, or by any other application dependent mechanism.

Our work focuses on protecting the database state, given a sequential input log of transactions.

A Byzantine-fault tolerant replicated state machine guarantees data integrity, even if a few replicas are corrupted by an attacker, as long as less than a small minority (usually less than 1/3rd) are corrupted. See [13] for a survey of work in this area.

The bulk of the work on BFT replication has focused on non-database applications. Most BFT replication solutions process the log serially in each replica. While serial processing is acceptable in some situations, for general database applications, serial execution has a very negative impact on throughput.

A number of papers have addressed Byzantine fault tolerance for data storage systems, but most of them do not support SQL, and none of them address efficient corruption detection and recovery, which is very important for large databases. Basil [45] supports Byzantine fault tolerance, but on a key-value store that supports limited operations, Callinicos [36] also supports key-value stores with limited operations (“mini transactions”) and does not support SQL, while Janus [60] addresses asynchronous BFT. Augustus [37] presents a scalable storage system, supporting partitioning and replication, which is robust in the presence of Byzantine failures, but has restrictions on transactions. None of the above systems support SQL, nor do they discuss corruption detection and recovery, which is the focus of our paper.

Early work in BFT replication for databases includes [21], and [23], but these too assumed serial execution. HRDB [52] allows concurrent execution on SQL databases. It uses a primary database to execute transactions concurrently, and define a transaction ordering and forces replicas to follow the same order. HRDB depends on strict 2PL, whereas current generation databases often use (serializable) snapshot isolation. Subsequent work on BFT database replication includes Byzantium [38] and Mitra [29]. Byzantium executes transactions in three parts, begin, execution and commit. Begin and commit operations execute serially in the log order. Transaction execution can run concurrently, on a snapshot determined by the transaction start. Mitra [29] allows a coordinator to run transactions concurrently first, and then replicas follow the same order. Both these approaches support SQL queries. Pesto [46] extends the ideas behind Basil to support SQL, but requires significant changes to the database. However, none of the above mentioned approaches addresses corruption detection or recovery.

HyperLedger Fabric [4] uses an execute-order paradigm, where transactions are “simulated” on multiple servers (endorsers), and if the endorsers agree on the read and write sets, the transaction gets committed. The most closely related work to ours is [33], which builds a “blockchain relational database” on top of PostgreSQL. Our transaction model is similar to theirs. They support order-then-execute and execute-then-order models; we follow the first model. But neither [4] nor [33] considers corruption detection and recovery.

chainifyDB [42] addresses BFT on top of existing SQL databases. As also mentioned in [42], earlier work in this space does not address recovery, whereas chainifyDB does pay attention to recovery. However, recovery is based on database snapshots, which are very impractical for large databases both for creation and for copying over a network. (Their experiments were on 1,000,000 records which is relatively small and still takes more than 800ms.) In contrast, we use Merkle trees for efficient recovery. Also, they detect corruption when the digest produced by the update transactions differs across

organizations. This detection may happen after very long time if corresponding data is rarely referenced. This also makes recovery expensive, as a lot of checkpoint snapshots would be needed. We detect corruption much faster by publishing the state of the database as merkle root hashes, which reduces need for numerous snapshots. After certain time, the snapshots will fill available memory and must be freed. If corrupted data is accessed after this time duration, the system will not have a healthy checkpoint snapshot to revert to. Our proposed solution easily overcomes this situation by replicating the state from another healthy replica.

Furthermore, they use triggers for tracking changes; an intruder who can modify the database can equally well disable triggers when modifying the database. In contrast, we use index update mechanisms to maintain Merkle trees, avoiding this vulnerability.

2.3 Database Recovery

When a replica crashes (possibly due to hardware issue or network disconnection etc) or acts byzantine (corrupted state), it needs to be detected and the state needs to be restored so that it can rejoin the cluster.

ARIES [31] is widely used to recover crashed relational databases. It restores state from checkpoint and then applies undo, redo operations to return database to functional state. Recovery can be done faster by reducing logging overhead (before/after images) and logging transaction commands instead, especially when failures are quite infrequent. [30] restores database state from previous checkpoint and then replays the transaction commands.

Various replication techniques are discussed in [22], [51] like (a) active model or state-machine replication that we use and (b) passive or primary-backup and (c) chain replication.

Multiple recovery alternatives are discussed in [25] such as (a) lazy replication to reduce overhead on functional replicas and break down the recovery of a replica outdated from long time, into incremental phases (b) full vs incremental data copy, and (c) reconstruction table proposal which tracks ID of last transaction that updates an entry. In case of partitioned database [24], leader replica of each partition may be different, which executes transactions with strict 2PL and recovery may be performed independently for each partition.

2.4 Deterministic Databases

The area of deterministic databases focuses on execution of transactions in a way that the serialization order respects the given input order, while supporting concurrent execution to improve throughput.

Early work in the deterministic database area included Calvin [48], and [39]. An overview of work on deterministic databases is provided in [2]. These papers address deterministic execution on a single node as well as deterministic execution in a partitioned-parallel database system.

Aria [28] is a deterministic execution protocol, where multiple transactions are batched and executed concurrently, on the same database snapshot. If a conflict between read-write sets is detected within a batch, the lower ID transaction is allowed to commit. Conflicting higher ID transactions are run again in the next batch,

on a newer snapshot. The order is guaranteed to be the same on all replicas, although it may not match the given input order.

Harmony [26] is another deterministic system that also executes multiple transactions in a batch concurrently for better performance [57]. Harmony also reorders conflicting transactions by re-executing them in the next batch. The conflict detection mechanisms in Aria and Harmony handle predicate reads also. The authors of Harmony have developed an implementation of conflict detection as done in Aria, by localized modifications to the PostgreSQL code.¹ We use this conflict detection code in our implementation.

Data corruption that does not show visible symptoms can be detected by comparing the state across hosts [54]. They create incremental snapshot with modified key-values but do not provide solution to recover.

2.5 Other Related Work

Ledger databases such as [55] provide verification and auditing support, preserving historical information for forward integrity checking [6]. [55] does not support SQL, whereas SQL Ledger [5] provides forward integrity for updates on an SQL database using a blockchain. However, these systems require an audit process for detecting corruption, and meanwhile wrong answers may be returned. Further they do not provide any efficient means for recovery from corruption.

Protection of data in outsourced databases has received much attention in the past. Here, the model is that there is only one copy of the database, with data integrity verified by means of digital signatures, and query result correctness checked by sending integrity proofs or “verification objects” with the query results, which are then verified by the client. While there has been a lot of research in this area, e.g. [10], [32], among many others, all the existing results in this area work only for very limited query types, and are also very expensive, making them impractical, unlike BFT replicated state machine approaches such as ours.

TAPIR [58] provides transactional guarantees on top of inconsistent replication. However, it does not deal with Byzantine failure.

3 System Model

Figure 1 shows our system model pictorially. In our model, clients submit transactions using digitally signed messages, which are added to a shared log, created using a consensus mechanism or other mechanisms.

Each replica in our Byzantine fault tolerance state machine replication model is implemented as an application server that provides a wrapper around a database system. The wrapper layer fetches each transaction from the shared log and processes the transaction. It performs authentication (using the signature), authorization checks, and then executes the transaction. Multiple threads are used to allow parallel execution of transactions, with the deterministic database component responsible for ensuring ordering. The results of transaction execution at each replica are sent back asynchronously to the client using Kafka’s pub-sub mechanism, to ensure fault tolerance.

We describe the components of the model in this section.

¹Available at <https://github.com/zllai/AriaBC>.

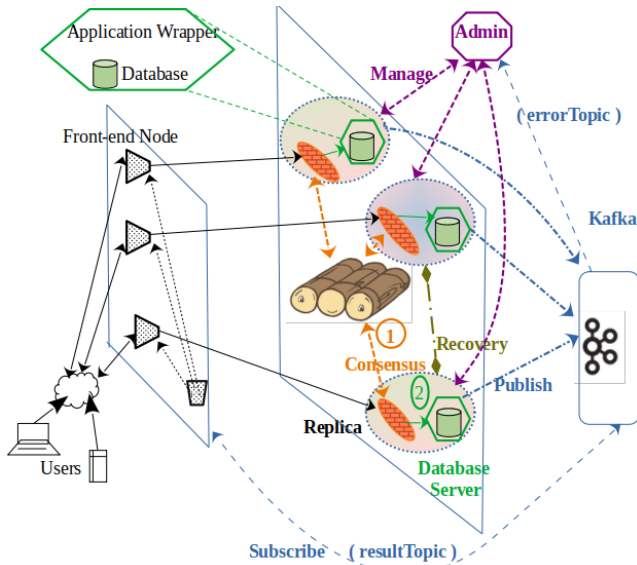


Figure 1: System with 2 healthy and 1 corrupt replica

Request	Client-ID	Sequence-Number	Transaction Statements	RSA-Signature
---------	-----------	-----------------	------------------------	---------------

Figure 2: Request format

3.1 User Interaction Model

Users submit transactions through a front-end client, which could be either their own device or through a web application. We assume for simplicity that clients are trusted and not compromised. Even if they are compromised, the damage is limited to operations within the authorization scope of the specific user.

As shown in Figure 2, each transaction contains (i) the user id, (ii) a sequence number that is unique for that user (to avoid duplication/replay attacks), (iii) the users transaction, which identifies the operation to be executed at the application server, along with its parameters, and (iv) a digital signature of the message contents.

Each front-end client also subscribes to Kafka to learn about the result of execution of the transaction at each replica. The client compares results from all replicas; as long as a majority of the full set of servers have sent the same value, it picks that value. Under the assumption that a majority of replicas are alive, accessible, and not corrupted, there would always be such a majority value, ensuring the system makes progress and does not block.

We discuss in Section 5.2 how to handle and resolve discrepancies if the results from different replicas differ. The comparison of results at the client can be optimized, by just sending a hash from all replicas except for one replica that sends the full result, and comparing the hash values and validating the result against the hash.

3.2 Transaction Log

Multiple users can simultaneously submit requests, with each containing one or more transactions. To ensure all replicas handle user requests in the same order, it is necessary that all replicas agree on the order of incoming requests. Thus, requests have to be added to a sequential log, which should be immutable for security reasons.

The input log can be generated by a Byzantine-fault tolerant consensus mechanism such as those used in (permissioned) blockchains, for e.g. [56]. If attacks on the consensus are not an issue of concern, and the focus is only on database state corruption, crash fault-tolerant consensus mechanisms like Raft [34] can be used, even though they are not Byzantine-fault tolerant.

There are also applications where the log is generated by an existing centralized system, but the state must be protected from attacks. For example, a stock market matching engine generates a sequential log of trades, which must then be processed by a depository which tracks who owns which shares; protecting the depository state from attack is a critical sub-problem. Where multiple parties are involved in a transaction, verification of consistency of such logs is done by periodically comparing the logs generated at the different systems, to detect and fix any inconsistencies, a process referred to as “reconciliation”.

Clients submit transactions to a frontend client, which then sends the transactions to the replica server. Servers run the consensus protocol to create the shared log. Once a transaction is successfully committed in the log, it is executed on the replicas.

If we used blockchain-based mechanisms, the use of larger blocks can cause significant delays in transaction execution. With permissioned block chains, it is possible to use blocks containing only one or a few transactions, to avoid latency.

3.3 Transaction Processing at Replicas

When the replica server detects that a transaction is committed in the log, it executes the transaction by sending it to the application server. We have an application server running on the same node as each replica server; thus, each server sends the transaction to its local application server.

The application server first validates the digital signature, and performs authorization checks to determine if the user is authorized to execute the transaction contained in the request. If the checks are passed the transaction is executed in the application server, on the underlying database. Otherwise, the transaction is deemed to have failed.

Transaction execution in our model has to be deterministic, with the serialization order identical to the order in the log. We use deterministic database execution techniques to allow concurrent execution while ensuring deterministic results, as described later.

Once the transaction is executed, the application server sends the results to the client, but indirectly via a Kafka pub-sub system to handle any crashes. The results are signed by the server, and the client can authenticate the signature.

Each of the replicas will receive transactions in the same input order. Thus, after execution, the database state of all replicas will be identical, provided the initial state was the same, and the majority of replicas are not compromised. We will see in a later section how the corrupted replicas are handled and restored.

It is possible to use the database itself as an application server, but then transactions would be restricted to stored procedures supported by the database.

3.4 Detecting Data Corruption

Participating server replicas that are corrupted or compromised may deliver different query results than the healthy majority, which will provide identical results.

Clients compare the results received from all the replicas. As long as a majority of the complete set of replicas have sent the same value, it is accepted, under the assumption that a majority of replicas are up and not corrupted. As long as only a minority of replicas are down or corrupted, the system will be able to make progress, else it may block. We discuss in Section 5.2 how to resolve discrepancies if the result hashes from different replicas differ.

As an optimization, only one of the replicas, chosen as the leader, can send the full result to the client, and others can simply send the result hash. The client can then compare the hashes with the locally verified hash of the full result from the leader. In case of any discrepancy, as long as a majority of the replicas sent the same hash value, the client can request one or more of the majority replicas to send the full result.

Update transactions are processed similarly to queries. The result of update SQL queries is usually very small, indicating the status of the operation and the number of rows updated.

In general, a user transaction executed in the application wrapper layer can be a combination of query and update operation. The application wrapper has the final choice in what result to send back to the client.

Our focus is on server compromise leading to data corruption at the server. However, it is also possible for a client to have been compromised. We note that if the client’s secret key has been leaked, any transaction sent ostensibly by that client would get validated. However, since these clients are typically end users who are authorized to only update their own data, the impact of such compromise would be limited to that client, thus minimizing risk to the overall system.

3.5 Key Management

Each client digitally signs the transaction with its private key. Similarly, each replica server digitally signs the execution result with its private key. Public keys of both sides must be stored in the recipients’ database or an external safe store. Any change in the public key should be done either with a certificate or through a trusted symmetric channel.

User keys being stored in the database, any update to keys is also a transaction on the blockchain. Thus, if a request signed with the old key arrives after the key-update request in the blockchain, it will be rejected.

4 Deterministic Execution Algorithm

Several schemes have been developed for deterministic execution, which allow concurrent execution; see e.g. [1]. These schemes typically allow transactions to execute concurrently, but commit serially, with transaction T_i restarted if a concurrent transaction T_j that is serialized earlier updated data that T_i read.

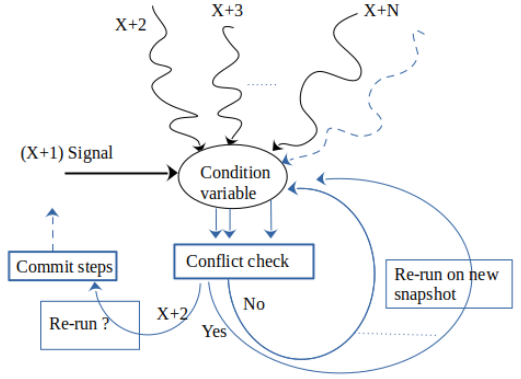


Figure 3: Workers and condition variable

Algorithm 1 gives the pseudocode of the deterministic execution algorithm that we use, which is based on Aria [28].

Each worker thread picks the next transaction (*current_id*) from the input log and executes the *RUN_Tx* function on the transaction. The function atomically performs two operations: (a) takes the current database snapshot S and (b) records the lowest unfinished transaction-id (*check_from_id* as the last committed transaction ID + 1 (Lines 25 – 26). (We assume that transactions are numbered sequentially in log/commit order.) Executing atomically ensures that all updates from transaction-IDs prior to *check_from_id* are committed in the snapshot. Thus, the current transaction needs to be checked for any possible conflict with transactions that had started but had not finished, before this transaction started.

The rw-conflict check is implemented as described in Aria [28], using code from an implementation of AriaBC on PostgreSQL from github.com/lzllai/AriaBC. As noted in [28], the read-write check also detects phantom conflicts since PostgreSQL uses leaf conflicts for checking conflicts between predicates and updates/deletes/inserts, and these conflicts show up as read-write conflicts on index leaves. We also note that the changes to PostgreSQL above are minimal.

The worker executes the transaction on the database snapshot S , to determine the read-write-set which was not known prior to scheduling the transaction (Line 26). However, it does not commit updates to the database, similar to [26], [28]. A transaction commits updates after verifying that it does not have any data conflicts with the preceding transactions (Line 30).

If there is a conflict the transaction is re-executed by the while loop, after a waiting time as determined in Line 28. The intuition behind the condition variable wait and conflict checking code in the while loop above is depicted pictorially in Figure 3.

If all preceding transactions have committed before snapshot S was taken (the check in the while loop in Line 27), the transaction will commit updates and send a signal to subsequent transactions (higher IDs) that may be waiting (Lines 35 – 36). Otherwise, if at least one preceding transaction had not committed before snapshot S was taken, this transaction will wait for the commit signal. This wait can be either for a certain threshold period of time, or a count of commit signals, etc (Line 28). The commit signal indicates the availability of a read-write set from a preceding transaction that committed.

Algorithm 1 Deterministic Transaction Execution Algorithm

```

1: RW_SET[] setA, setB           ▶ global read-write hash table sets
2: RW_SET mapToTable(Tx_id);     ▶ based on switchThreshold
3: last_committed_tx_id ← 0      ▶ global across threads
4: compare_from_id ← 0          ▶ private to worker
5:
6: function COMMIT (Tx_idi)
7:   currentSet ← mapToTable(Tx_idi)
8:   (currentSet != mapToTable(Tx_idi-1)) ? empty(currentSet) : NO-OP
9:   Insert (i, rw_seti) into currentSet
10:  last_committed_tx_id ← Tx_idi
11: end function
12:
13: function CHECKCONFLICT (TX_ID1, TX_ID2, RW_SETi)
14:   for ( i ← TX_ID1 ; (i ≠ TX_ID2) ; i ← i + 1 ) do
15:     currentSet ← mapToTable ( i )
16:     if ( (RW_SETi conflicts with RW_SETj from currentSet) &
17:           (TX_IDj ≥ compare_from_id) ) then return true
18:   end if
19: end for
20: return false                  ▶ No conflicts
21: end function
22: function RUN_Tx (Tk) :
23:   for ( loop ← true ; (loop) ; ) do
24:     oldest_commit_id ← last_committed_tx_id
25:     compare_from_id ← oldest_commit_id + 1
26:     rw_setk ← Execute(Tk, DB_Snapshot() )
27:     while (oldest_commit_id ≠ (k - 1)) do
28:       WaitForCommitSignals() ▶ e.g. periodic, count-based etc.
29:       latest_commit_id ← last_committed_tx_id
30:       loop ← CheckConflict( oldest_commit_id,
31:                             latest_commit_id, rw_setk)
32:       oldest_commit_id ← latest_commit_id
33:       ( loop ) ? break : continue
34:     end while
35:   end for
36:   Commit(Tk)                  ▶ Publish read-write sets
37:   CommitSignalBroadcast()      ▶ Signal waiting workers
38: end function

```

When a commit signal is received, the worker will (a) record the ID of that transaction and (b) compare the read-write set of the current transaction with all those who have published a commit signal (i.e. committed, *Line 30*). In the optimized case, this translates to the number of signals reaching threshold count or timer expiry. It then compares the read-write sets of all transactions till this recorded ID, with its own read-write set (*Line 13*).

If there is a conflict, the transaction is restarted by taking a new database snapshot with the latest state. If there is no conflict, and all conflict checks with preceding transactions are done, a transaction can commit (*Line 6*). It commits its changes and saves read-write sets to the global hash table (*Line 7 – 10*) and then sends a broadcast signal to other workers (*Line 37*).

The hash table stores read-write sets of all committed transactions. However, once a transaction is old enough that no currently executing transaction is concurrent with it, its read-write sets can be deleted.

To simplify deletion of data from the hash table, we use a pair of hash tables setA and setB, where one table is used until it fills up, and then the other table is used until it fills up. The function mapToTable() (*line 2*) determines which table to use for the current transaction, based on the transaction ID number. The tables are sized based on the maximum number of threads N , such that when setB becomes full, all current transactions in setA are redundant, which can be deleted, and symmetrically for setA becoming full. Due to commit ordering, with N threads the oldest currently executing transaction cannot be more than N transactions before the newest currently executing one.

It is possible that upon re-execution on a new snapshot, the transaction may follow a different codepath from the original simulated execution, resulting in new read-write sets. This is expected to be a rare scenario, and even in this case, the new execution will account for all conflicts.

5 Corruption Detection & Recovery

In this section, we discuss different threat models and the detection of a corrupt replica passively and actively. We present a mechanism to update the Merkle tree with each transaction, along with partitioning. Finally, we describe how to recover a compromised node and discuss integrity constraint violations during recovery.

5.1 Threat Models

There is a range of attacks possible depending on system vulnerability and the attacker's level of sophistication. We consider multiple threat models reflecting different levels of security breaches.

In the first threat model, we assume the database system code is not compromised. Even when an attacker has broken into the system and tampered with data, the database configuration and automatic update of the Merkle tree remain intact. On the compromised replica, the Merkle tree will generate a different root hash that does not match the state of a healthy replica. When root hashes of all replicas are compared, compromised replicas can be identified as long as they form a minority.

In the second threat model, an attacker silently corrupts the database by physically tampering the data files.

In the final threat model, which we call the major threat model, we consider that the database system code may also be compromised; thus, the Merkle tree hash may not match the underlying data, and the replica may fool a recovering replica into accepting wrong results. Yet, the result of the transaction provided by this replica based on tampered data will not match the majority result, thus indicating a security breach.

To handle this model, we need to first restore the code, then recompute Merkle trees, and then restart the protocol. The rest of our description focuses on the case where code has not been modified.

Activities such as user public key update are done by the system administrator. A compromise of the system administrator's private key could impact multiple users. We assume in this paper that the administrator credentials are not compromised.

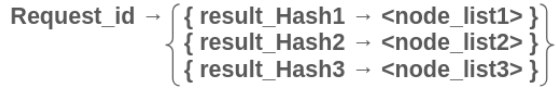


Figure 4: Transaction Result Aggregation Map

5.2 Corrupt Replica Detection

All replicas receive an identical sequence of globally ordered transactions. In passive detection, the result of each transaction is monitored. If the result from any two database replicas differs, at least one of them must be corrupted. Since only a minority of replicas may be compromised, we can determine the correct transaction result based on the majority. However, data corruption may manifest in transaction result mismatch after a very long time. Hence, a system administrator may want to detect data corruption early, rather than passively waiting for it. We propose a passive and an active scheme below, and later present a mechanism to recover a corrupt replica.

5.2.1 Passive Detection Using Transaction Result . When the result of a transaction execution is found to be different on different replicas, we can infer that one or more replicas have been corrupted.

Each front-end client maintains a data structure shown in Figure 4, which stores for each transaction (request), the set of result hashes received from the replicas, along with the list of replicas that returned that value. This data structure is used to determine the correct value (returned by a majority of the replicas in the system). All replicas whose result hashes differ from the majority are detected as having been corrupted.

This passive detection scheme only detects corruption that affects the result of a query. If some data that has been corrupted is not queried for a long time, that corruption would correspondingly not be detected until it is queried.

5.2.2 Active Detection using Merkle Tree . In the active detection scheme, the database state at all replicas at corresponding points in the input log are compared with each other, by using a Merkle tree whose root node hash represents the state of the entire database. The database state is obtained by taking a database snapshot, and all queries, including queries on Merkle tree data, run on this snapshot. A textbook description of Merkle trees for such comparison may be found in, e.g., [44]

To actively detect a corrupt replica, a special transaction, 'compareStates', is inserted into the blockchain to gather database state at all replicas at the same instant in the transaction log. When a compareStates transaction is found, the application server at the replica does the following atomically: (a) it saves a database snapshot to be used later during the recovery procedure and (b) it publishes the state, represented by the root hash of the Merkle tree (or root hashes of all trees in the partitioned Merkle tree case) of each relation. These states are then compared, and any divergence is used to detect corrupt replicas.

Such active detection can be performed either periodically by the system or initiated on demand by system administrator.

Some database replicas may be unavailable or lagging in execution, and the root hash at that instant may not be available within a reasonable time. Hence, comparison of the rest of the replicas

can be done without it, possibly after a timeout. As long as healthy replicas form the majority, we can check such lagging replicas when the root hash at that point in time becomes available.

5.2.3 Merkle Tree Update. Each row in a user relation is mapped to a leaf node in the Merkle tree using a hash function. Each time a transaction updates a row, the corresponding Merkle tree leaf value is updated. This leaf update involves removing the contribution of the previous data hash and incorporating the new data hash. This updated value at the leaf node triggers an update to its parent node value, which is done similarly, by removing the contribution of the old leaf hash and incorporating the new leaf hash. This updating of parent node value continues recursively up the tree till the root node. All these updates of Merkle tree nodes are done as part of the original transaction itself, which updated the row.

We note that the Merkle tree nodes are maintained as records of a system relation, ensuring existing conflict detection mechanisms also handle conflicts on the Merkle trees.

Normally, two concurrent transactions updating distinct rows in a user-defined relation can execute independently by locking their respective write-sets.

If the modified rows map to the same Merkle tree leaf, write-sets will imply conflict at all nodes from leaf to root. Even if the modified rows map to different Merkle tree leaf nodes, both transactions will update all parent nodes, up to the Merkle tree root. Thus, write-sets of such transactions will not be disjoint if both impact a common Merkle tree.

Conflicts in updating common items in write-sets, in particular a common root node, would severely limit how many transactions can be processed simultaneously. We therefore use a partitioned Merkle tree.

Partitioned Merkle Tree: To avoid Merkle tree update from being a bottleneck for concurrency, we logically partition each relation and compute a (smaller) Merkle tree on each partition [20]. A transaction updating a Merkle tree leaf now causes updates to its parent nodes only till the root node of its partition. Thus, write-set conflicts between transactions when updating the Merkle tree are significantly reduced.

Figure 5 shows a partitioned tree, with our node ID assignment scheme that keeps node IDs unique across all partitions. Node hash data of all nodes across all partitions of the tree is maintained in a single relation, whose rows are identified by the node ID scheme shown.

Transactions modifying user data that map to leaf nodes in the same partition may still conflict while updating Merkle tree nodes, but the partitioning scheme and count can be chosen such that conflicts are greatly reduced. Transactions updating data in different partitions will have no common Merkle tree node, and such updates can be applied independently. On the other hand, a transaction updating data across multiple partitions will limit the concurrency gain offered by partitioning. In all cases, the transactions commit in the same global input log order defined by the blockchain.

When the content of a database replica is compromised, as identified by a difference from other replicas in either Merkle tree root hash or a transaction result, that indicates the need for a recovery mechanism to bring the replica back to operational state.

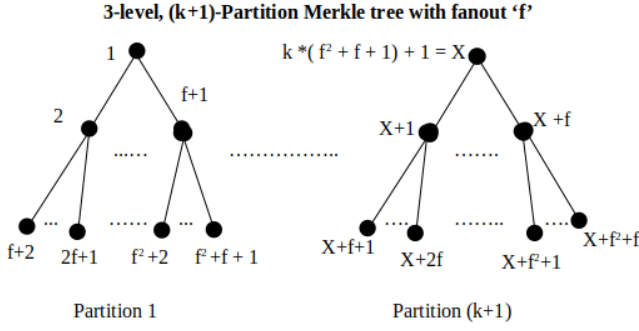


Figure 5: 3-level, (k+1) Partitioned Merkle tree with node-IDs

Algorithm 2 Recovery Algorithm

```

1: procedure RECOVERY (REPLICA  $N_k$ )
2:                                      $\triangleright$  active detection start
3:   mark_log_offset  $\leftarrow$  current_log_offset ('compareStates' transaction)
4:   For each replica  $N_i$ :
5:      $S_i \leftarrow DB\_Snapshot(replica\ N_i, mark\_log\_offset)$ 
6:      $M_i \leftarrow Merkle\_Tree(S_i)$ 
7:   [ $G_{majority}, G_{rest}$ ]  $\leftarrow Group\_replicas\_by\_merkle\_root\_hash()$ 
8:                                      $\triangleright$  active detection completed
9:                                      $\triangleright$  recovery phase start
10:  ( $N_{ref}, M_{ref}$ )  $\leftarrow$  reference instance of ( $replica\_ids, G_{majority}$ )
11:  Compare_merkle_trees_and_Replicate_rows( $M_k, M_{ref}, N_k, N_{ref}$ )
12:  Replay_input_tx_logs_and_rejoin_cluster(mark_log_offset)
13:                                      $\triangleright$  recovery phase complete
14: end procedure

```

5.3 Corrupt Replica Recovery

The state of a database replica may be corrupted by malicious updates. System failures, like power or network, or component failure, which result in a replica falling substantially behind the rest of the system can also be treated similar to corruption, to recover the state, without playing back a very long transaction log.

Before initiating recovery, we first stop processing at the corrupted replica and ensure that the intruder no longer has access to the system. With manual intervention, the security loophole on the corrupted replica is fixed, like a security patch or system update, to prevent it from being compromised again in the future.

We describe the procedure performed by the system administrator to recover a corrupted replica pictorially in Figure 6, and with pseudocode in Algorithm 2.

The first step is to identify a healthy replica to use as a reference. To do so, a logical snapshot S_1 of the current database state is taken at all replicas (Line 4,5), as of a certain point in the transaction log as explained in Section 5.2.2. The logical snapshot S_1 includes the user data as well as corresponding Merkle trees. The default mechanism for taking snapshots in PostgreSQL takes care of this step.

We choose one of the majority of replicas which have an identical Merkle tree root, thus an identical database state, as the reference replica (Line 10).

Next, we identify the differences between the Merkle trees of the corrupted replica and the reference replica (Line 11) on the snapshot

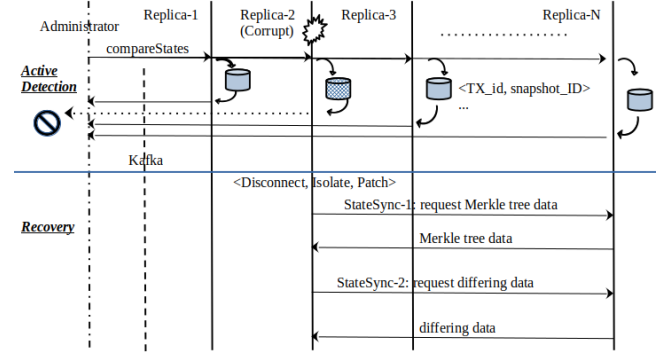


Figure 6: Procedure to recover a failed replica

S_1 taken in the earlier step. Since the root hash of the corrupted replica is different, the hash of at least one of the children of the root node must differ from that in the reference replica. By recursively comparing the hashes of children down the tree, we can identify leaf-nodes whose hashes do not match across replicas. In case of partitions, as proposed in the previous section, the tree comparison must be done between all corresponding partition-trees.

From mismatching leaf-nodes, we identify differing rows in the database by using an index mapping Merkle tree leaf nodes to the rows whose hash values map to the node, sometimes referred to as functional index. Such an index is easily implemented in PostgreSQL using a “function index” on the table, wherein an index is created with its key as a function that maps the row to the Merkle tree leaf identifier. The differing rows are then resolved by replicating the state of those rows from snapshot S_1 of a reference database replica.

At the end of the above phase the partition has been recovered up to the point in input transaction log when the snapshot was taken. The state of the erstwhile corrupted replica is then brought to the current state by replaying the input transaction log starting from just after the point in the log where the snapshot was taken, during the active detection procedure (Line 12).

It is possible to restore a minority of corrupted replicas concurrently with this procedure, even if they were corrupted at different times. Also, a healthy replica may serve as a reference for multiple corrupt replicas simultaneously.

Our mechanism utilizes snapshots, which most current generation databases support. In PostgreSQL, we can get a specific snapshot with an identifier, and then run multiple queries on that snapshot. We use this feature in our implementation.

5.3.1 Integrity Constraint Discussion. Note that the database state at the healthy replica was reached by executing transactions in the order they appeared on the blockchain. However, the recovery program imports all the differences from the healthy reference replica to the corrupt replica, without any order. Hence, while the recovery program is being run, the corrupt database instance may temporarily violate integrity constraints. For example, a relation P may refer to a foreign key in relation Q. A transaction may add new entries to both relations on a healthy replica. However, the recovery program may first update relation P entirely, while the foreign key entry in relation Q is yet to be updated.

Hence, constraint checks are disabled while the recovery is in progress and re-enabled when complete. After all differences are applied, the recovered replicas will satisfy all constraints, and integrity checks are re-enabled.

6 Dynamic Merkle Tree

As the database size increases with row count, more tuples map to a leaf node in the Merkle tree. Hence, during corrupt replica recovery, the time to fetch the tuples and compare them across the replicas increases linearly with the number of tuples mapping to a merkle leaf node. To reduce the recovery time of a corrupt replica, we propose to limit the number of tuples mapping to a leaf node, with a Dynamic Merkle tree data structure that scales dynamically as the user relation grows/shrinks.

While the structure of B+tree also limits number of keys per node and scales dynamically with input values, it is non deterministic. Thus, two replicas executing transactions in the same logical order, but with some non-conflicting operations ordered differently, can internally have two different B+tree structures for the same data. Thus it is not possible to extend a B+tree with node hashing to efficiently find differences between replicas.

We require the ability to compare replicas where tuple insertions may occur in different orders and thus, internal tree structure be determined only by the tree content, regardless of the order of operations. Our proposed Dynamic Merkle tree data structure preserves this property of determinism, while allowing the tree to grow or shrink incrementally ensuring that the number of tuples mapping to a leaf node is bounded. When the number of tuples mapping to a leaf node exceeds a certain threshold, our Dynamic Merkle tree is restructured by making that node as internal node and creating multiple children among which the mapping tuples are divided based on tuple hash.

Observe that depending on the skew in input data, some tree branch may grow deeper by adding child nodes more often and some other branch may shrink, resulting in a tree with non-uniform depth across branches from root to leaf. Thus, unlike a B-tree, a Dynamic Merkle tree is not a balanced data structure. However, as long as hashing functions are well designed, leaf depths will be relatively uniform.

Similarly, when tuples are deleted and the count of all tuples mapping to descendant nodes of an internal node falls below a certain threshold, the child leaf nodes can be merged into the parent node. Note that this merge-up can also happen recursively across multiple levels.

Another important requirement is the ability to save snapshots of the Merkle tree data structure, which are used for comparing the state across nodes. Since many databases already support snapshots of database data, we can reuse existing database snapshot support by storing the merkle tree as a relation, and is explained in subsection 6.2.

In subsection 6.1, we describe the data structures maintained within database to support resizing the merkle tree dynamically. In subsection 6.2, we describe in detail, the algorithms used to update merkle tree during insert, delete and update operations on user relation.

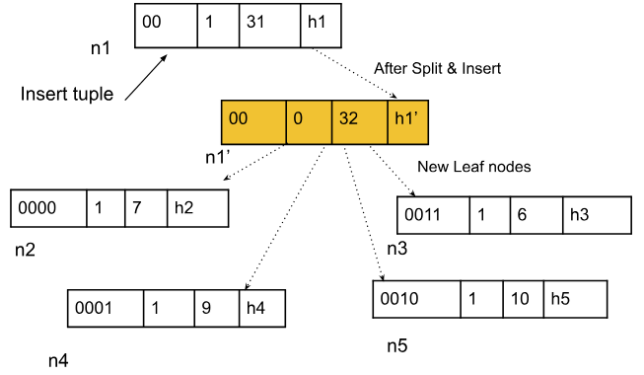


Figure 7: 4-way Leaf Node Split in Dynamic Merkle Tree

6.1 Dynamic merkle tree data structure

For each user relation R , a new relation R_merkle_nodes is created internally with an entry for each merkle tree node (both leaf and internal) having attributes as follows:

(*part_id*, *node_id*, *is_leaf*, *tuple_count*, *node_hash*)

Here, the first two columns indicate the merkle partition_id (*part_id*) and the *node_id* which is unique within the partition. An internal node has 2^k child nodes whose *node_ids* are generated by suffixing bit sequences of fixed length (k) to the parent *node_id*. For example, with $k = 2$, a parent node has 4 children, the *node_ids* for child nodes are generated by suffixing one of {00, 01, 10, 11} to the *node_id* of the parent node.

Note that the *node_ids* of child nodes of an internal node are mutually exclusive and completely span the universe defined by the *node_id* of the parent node. Thus, if a tuple mapped to the parent node, it will map to exactly one child node as only one child *node_id* will match the the bit-prefix of tuple_hash. Recursively performing this, we locate the leaf node to which the tuple maps. A tuple mapping to a leaf node is considered to be mapping to all parent nodes of the leaf node, upto root node. A *node_id* matches bit-prefix of the hash of all tuples mapping to that node. Thus, when a tuple is updated, its tuple_hash will change, and the updated tuple will map to another leaf node, most likely different from the original leaf node.

The value of the next attribute *is_leaf* depends on whether the node is internal (0) or leaf (1). The last two attributes indicate (a) the count of all tuples mapping to this node and (b) the aggregation of tuple hashes of all tuples mapping to this node respectively. A tuple hash is computed as the hash value of all the attributes of the row. A B+tree index is created on tuple_hash, which is created as a covering index with extra attribute of user relation key. Thus hash values are contiguous as we will see in section 6.2.1.

The merkle tree schema presented above remains the same for both static and dynamic merkle trees. For static merkle tree, the number of partitions, nodes and node-types (*is_leaf*) remains constant. In contrast for dynamic merkle tree, only the number of partitions remains constant. Leaf nodes with too many mapping tuples (large *tuple_count*) are split to create child leaf nodes thus changing node-type of original node to internal (*is_leaf* = false)

In both cases, *tuple_count* and *node_hash* change with tuple operations.

We explain in the next subsection 6.2, how the merkle tree *R_merkle_nodes*, is updated whenever a tuple is inserted, deleted or updated.

6.2 Dynamic merkle tree tuple operations

For all types of operations that modify a user relation tuple, the first step is a common procedure to identify the merkle tree leaf node to which tuple maps. For each tuple operation in user relation, we traverse the merkle tree down from the root node, at each step, finding the child node whose *node_id* is a prefix of the *tuple_hash*. At the root node, the prefix length used for comparison is say k (*BITS_PER_LEVEL*) and the length is incremented by the same amount k at each level down the tree. As discussed earlier, whenever a tuple in user relation is modified, the tuple hash determines the merkle leaf node the tuple maps to.

6.2.1 Dynamic merkle tree tuple insertion . When a new tuple is inserted in the user relation, the leaf node to which the tuple maps in the merkle tree, is determined as described above. For all nodes along the path from the root to the leaf, the entries in the *R_merkle_nodes* relation are updated by (a) incrementing the *tuple_count* and (b) including the *tuple_hash* in the *node_hash*.

When inserting a tuple in the user relation, if the merkle leaf node it maps to, has *tuple_count* equal to a certain threshold, it is split by creating new leaf nodes as its children, thus making the original leaf node, an internal node. The new child leaf nodes have *node_ids* which are k -bit longer than their parent, and are generated by concatenating parent *node_id* with each of (2^k) binary sequences. The tuples mapping to the original leaf node are distributed between its child leaf nodes based on which new *node_id* their tuple hash prefix matches. The B+tree index on *tuple_hash*, helps efficiently search for the tuple hashes mapping to a node in the merkle tree, since range search accesses contiguous memory, rather than random I/O.

As shown in Figure 7, the leaf node *n1* has 31 tuples mapping to it in a merkle tree configured with $k = 2$ and threshold of 31. For simplicity, common parameter *partition_id* is omitted, and is identical for all parent and child nodes. After the new tuple insertion, the maximum threshold leaf size would be exceeded, leading to node split. The leaf node *n1* (before split) becomes parent node *n1'* (shown shaded in figure) with 4 child leaf nodes (in this 4-way case) with node-ids 0000, 0001, 0010 and 0011 shown as nodes *n2*, *n3*, *n4*, *n5* respectively. The tuples are mapped to child nodes based on their hash prefix and may not be equally distributed among the new child leaf nodes. In the example, after tuple insertion, node *n1'* with *node_id* (00) matches prefix of all 32 tuple hashes, however, the number of tuple hash prefixes matching new leaf nodes are 7, 9, 10 and 6 respectively.

For newly created leaf nodes *n2*, *n3*, *n4* and *n5*, it is necessary to compute the hash values from the tuple hashes, since the earlier leaf node *n1* only had aggregated hash for all tuples. Note that the *node_hash* value of the node *n1* is only updated with hash of tuple being inserted, since the union of set of tuples mapping to its new child nodes is the same as the set of tuples that mapped to it before the split.

6.2.2 Dynamic merkle tree tuple deletion . Similar to insertion operation, when a tuple is deleted from the user relation, the leaf node to which the tuple maps in the merkle tree, is found as described above in subsection 6.2. For all nodes along the path from the root to the mapped leaf, the entries in the *R_merkle_nodes* relation are updated by (a) decrementing *tuple_count* and (b) excluding the *tuple_hash* from the *node_hash*.

Conversely to discussion of tuple insertion, we first check the parent of leaf node to which the deleted tuple mapped. If the number of tuples mapping to the parent node of that leaf node falls below a certain threshold, all the child leaf nodes are merged into parent node thereby reducing the height of that branch of the tree. The parent node (in *R_merkle_nodes* relation) is rewritten as a leaf_node by merging tuples at all the child nodes. The hash of the parent node needs to be recomputed only for the tuple being deleted, since all the other tuple hashes already mapped to it through the earlier child leaf nodes. This merging-up may be repeated for parent of the parent node up the tree, if *tuple_count* of all peers of the parent node (other children of parent of the parent node) is 0. This merging-up is performed recursively until we reach a parent node whose *tuple_count* is above the threshold. The example shown in Figure 7, can be viewed as an example for tuple deletion from user relation, by merkle tree node-merging with direction of arrows reversed.

6.2.3 Dynamic merkle tree tuple update . Similar to insertion and deletion operations, when a tuple in the user relation is updated, the leaf nodes to which old and new tuples map in the merkle tree, is found as described above in section 6.2. A tuple update operation results in merkle tree update equivalent to old tuple being deleted and new tuple being inserted.

Note that when both old and new *tuple_hashes* map to same merkle tree partition, the *tuple_count* will not change for any node which is common parent of leaf nodes to which the old and new tuple map. In the special case where both old and new *tuple_hashes* map to same leaf node, the *tuple_count* will not change for any node. Also, no change is needed in the merkle tree structure since no merkle tree nodes are created or deleted. In this case, it is possible to optimize the *node_hash* operations by combining into single operation if the hash-combining function is associative and distributive.

6.3 Recovery with Dynamic merkle tree

As previously mentioned, leaf nodes in a dynamic Merkle tree may split during insertions or merge during deletions when their tuple counts hit maximum or minimum thresholds respectively. Thus tuple operations when executed by an attacker on a compromised replica can result in merkle tree structure that differs from the structure in a healthy replica. When recovering such compromised replica, the recovery procedure must check if the merkle tree structure matches healthy replica, in which case the procedure described earlier will work.

Merkle tree structure mismatch across replicas can arise from four distinct cases: (a) insert into leaf node at compromised replica which becomes an internal node with children, whereas on healthy replica, corresponding node remains a leaf node or (b) insert into leaf node at healthy replica which becomes an internal node with

children, whereas corresponding node in compromised replica remains a leaf node or (c) delete from a leaf node at compromised replica which merges with one or more peer node(s) to shrink the tree, whereas on healthy replica, corresponding node remains a leaf node or (d) delete from a leaf node at healthy replica which merges with one or more peer node(s) to shrink the tree, whereas corresponding node in compromised replica remains a leaf node. We now describe how to recover a compromised replica with merkle tree mismatch with healthy replica by one-level.

Recall that merkle trees are compared across replicas starting with root node and recursively finding the child nodes whose hash values differ. In case of merkle tree structure mismatch, we perform such comparison recursively until either one of the differing merkle node is a leaf. We then determine the set of keys mapping to the leaf node on one replica and compare it with union of keys mapping to all descendent nodes in other replica. Since the internal nodes only maintain `tuple_count`, it is necessary to find all the child nodes and their corresponding keys. If the healthy replica has shorter tree, the excess keys on compromised replica are purged from the merkle tree. Conversely, if the compromised replica has shorter tree, the missing keys found from the union of keys on healthy replica are inserted. Note that this must be done for each path from root to leaf node, that differs across replicas.

Note that as replicas execute more operations, such discrepancies may compound and the merkle tree structures may exhibit multi-level mismatch. Since our corruption detection mechanism is nearly real-time (within end-to-end system latency), the structures are not likely to deviate too far. The procedure described above for one-level merkle tree mismatch can be extended for multi-level mismatch by considering the union of keys mapping to multi-level descendants.

7 Handling Non-Deterministic Transactions

The architecture of our system assumes that transactions are deterministic. However, a badly written transaction may be non-deterministic; for example, the execution result may depend on a random number, or dynamic system-parameters like time, thread-id, etc. In this case, despite executing transactions in the same deterministic order across all replicas, database states across replicas could diverge. This is different from database corruption due to a security breach, which was discussed earlier, and could result in each replica returning a different result, or having a different state after the transaction.

Our system already compares the query results from different replicas, and will detect the case where the query result is non-deterministic. However, transactions may update the database non-deterministically, resulting in divergent states across replicas, while returning a deterministic query result.

We present below detection and recovery mechanisms for such non-deterministic transactions.

7.1 Detecting Nondeterminism

To detect database state mismatch across replicas due to a non-deterministic transaction, we propose a simple extension of the scheme used earlier for passive detection of data corruption. Writes to the database only return an execution status, and even if the

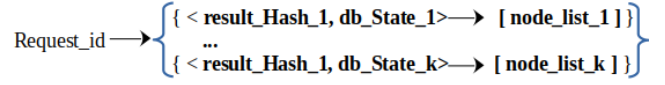


Figure 8: Transaction Result and State Aggregation Map

actual writes are different in different replicas, the status may be identical.

To detect divergence, in addition to returning transaction results, each replica now also sends the updated Merkle tree root hash of all the partitions that are updated, for each updated relation. The partition root hashes are compared, along with result hashes, to detect divergence.

Figure 8 shows an extension of the Figure 4 for detecting database state divergence. The client collects the result hash and updated partition hash information (i.e. partition ID and root hash) for all partitions updated by the transaction.

We would like to highlight that this mismatch is detected immediately after the commit of a nondeterministic transaction. Note that this extended scheme can also be used to improve accuracy of passive corruption detection described earlier, in Section 5.2.1.

The above detection mechanism is likely to be particularly useful for debugging transactions, since it catches non-determinism immediately, allowing easy detection of the cause for non-determinism.

In a well designed system there should ideally be no occurrence of non-determinism, but the check could be done anyway since it is very low overhead, similar to `assert()` function calls in programs.

However, in case non-deterministic execution does happen in a production system it is important to be able to recover quickly from it. We describe how to do so next.

7.2 Recovery process

When a database state divergence is detected across replicas, it could be either because of a security breach or non-determinism. While the underlying cause for the non-deterministic execution of the transaction needs to be fixed, if it was a rare error, example due to a “heisenbug”, there may not be another occurrence for a while, giving time to fix the bug. A repeated error would, however, cause slowdown of the system due to recovery, and need urgent fixing.

As long as a majority of the replicas give the same result, the recovery method described in Section 5.3 can be used without any change. However, it is possible that there is no majority group. In that case, recovery can be initiated after deciding which replica to use as the correct copy, which can be done by either choosing one which is in a majority, or if there is no majority, by a simple majority, or by decision of a system administrator.

The recovery mechanism proposed earlier for corruption can be applied here with minor enhancements as described below. The key differences in the recovery mechanism are (a) how reference replica is chosen based on Merkle root hash (or in case of partitioned tree, partition root hash) values across replicas and (b) recovery may be done for multiple replicas (in the limiting case, for all replicas but one).

In case some replicas report common states, we could use the set of replicas with highest cardinality (if one exists) as reference. As observed earlier, in the extreme case, each replica may give a

different root hash. In this case, there is no single source of truth and we can choose any one as reference. Those replicas whose database states are different from reference replicas, are referred hereafter as diverged replicas.

A special transaction 'saveSyncState' is inserted into the shared input log, listing all reference replicas. It causes each replica to save database snapshot and check if it is listed as reference replica. If a replica is not a reference, it needs to stop processing and recover state from one of the reference replica listed, and after that it can fetch newer requests from the shared input log. The replica listed as reference can simply save the snapshot, and continue to process newer requests without interruption.

Note that non-determinism is detected after transactions have committed at the replicas. Transactions added to the shared input log before the special recovery transaction is inserted will be processed by the replicas, and some of these may have a dependency on the non-deterministic transaction. Since our detection mechanism is quick and no manual intervention is necessary, there will be very few such transactions. Database state changes due to these transactions, as well as those due to the non-deterministic transaction, are replicated from a reference replica and would be consistent after recovery. But results returned to users based on these transactions could potentially have errors.

If a transaction does not depend on the non-deterministic update, the majority would have returned the same result as the reference value. If it did, and there is a majority with the same value for the non-deterministic result, the dependent transaction would have returned the same result on all these replicas. If there is no majority, the dependent results would also not be in a majority except in an unlikely scenario where divergent inputs gave the same transaction result for the dependent transaction. In this unlikely case alone we may need to inform a user that the value returned may changed.

8 Scalability

We now discuss how our proposed model can be scaled when set of replicas change (Section 8.1) or for heavy workloads (Section ??).

8.1 Reconfiguration

Whenever set of replicas participating in the system changes (addition or deletion), the administrator needs to communicate this update to all other replicas, so that all the group decisions like consensus, leader election, majority determination are done correctly with new set of participating replicas. When one or more replicas are removed, this update is sent immediately. However, when a new node is added to the system, it needs to have the latest state before it can participate in group activities or act as a healthy reference node to recover another corrupted node.

When a replica is newly added, the administrator first inserts the checkpoint transaction (as described in section 5.3) into input log, so all replicas will save a snapshot of current database state at a marked offset in the input log. The administrator then assigns a healthy reference replica to synchronize the database state to the newly added replica. Once the new replica has synchronized with the state of healthy replicas at the marked offset, it can then execute the transactions from the input log. As long as the transaction processing rate of the new replica is higher than the transaction

input rate, the input log offsets being processed at new and healthy replicas will converge. Eventually, the transaction results from the new replica will be published before the timeout window for the majority determination of the transaction result. At this instant, the administrator will publish the update to include the newly added replica in majority determination (and possibly a new majority threshold). This can be done by publishing the update as a transaction in input log so that all replicas remain in consistent state and execute the update deterministically.

The newly added replica can either participate in input consensus (by spawning a new instance) and receive the log or continue to receive input log from another healthy replica.

9 Implementation Details

In this section, we describe several implementation details.

9.1 Transaction flow through System

Transaction requests are sent by user clients to front-end nodes. Requests arriving at these nodes are sent to a Raft server, which uses consensus with other server instances to determine a shared order, in a crash fault tolerant manner. We use Ratis [19], which is an open source implementation of Raft. In our implementation, each Raft server replica has a unique ID and is configured and initialized identically with a Raft server instance, an application server, and a PostgreSQL database. The Raft server instance on each node sends transactions that have reached commit status on the Raft log to the application server. The server is multi-threaded, allowing concurrent execution of multiple transactions (deterministic ordering is handled at the database side). Database requests are sent to the local PostgreSQL database server through the connection pool. The number of threads and size of the pool is configured to be the same as the number of workers on the PostgreSQL server.

The application server receives query results from the PostgreSQL server, and on completing the transaction, publishes the transaction result to Apache Kafka result topic [18] along with its server ID . During corruption recovery, Administrator connects to application server to (a) initiate Merkle tree comparison (b) lookup data items using functional index and (c) replay transaction log.

Kafka service has separate topics to hold transaction results, and server errors, respectively. The Ratis client which received a user transaction gathers results for the transaction from all replica servers via the Kafka result topic and stores them in a hash-map as described earlier. If the client detects a divergence in results from one or more replicas it publishes a message to a Kafka error topic to trigger recovery actions by the Administrator.

9.2 Database processing

Database executes transactions deterministically as per our algorithm, which is based on the implementation of Aria [28] in Postgres, provided by the authors of HarmonyBC [26].

9.2.1 Initialization . We create schema-specific user relations, functions on each Postgres database replica, with all constraints as deferrable . For each user relation, we create a Merkle tree, implemented as an index. All relations and Merkle trees are set to the same initial state across replicas.

Rows in a user relation are mapped to Merkle relation leaf nodes uniformly with a hash function. The hash of each user data row is computed by hashing attribute values with Blake3 [35]. Hashes of all rows mapping to a leaf node are combined using the XOR function to allow efficient incremental update of the leaf hash values. Hash-value of a parent node is computed by combining the hash values of all child nodes again using the XOR function. The Merkle tree is configured identically on all replicas (schema, number of leaf nodes, fanout, mapping and hashing functions, partitions, etc).

9.2.2 Merkle Tree. We created a new index type in PostgreSQL to maintain the Merkle trees. For each operation done on the user relation by a transaction, the index update is invoked automatically by the database code as apart of the user transaction itself. The index implementation updates the rows of the Merkle tree relation corresponding to leaf node to which tuple maps and its parents up through the root. Normal concurrency control and recovery techniques get applied on the relation representing the Merkle tree.

For partitioned Merkle trees, one relation holds the hash values of Merkle tree nodes of all partitions, each with a different ID range. With P partitions, fanout of F and k levels in the Merkle tree, there will be P root nodes, and node count $N = (F^{k+1} - 1) / (F - 1)$ per partition and $P * F^k$ total leaf nodes and Merkle relation with $P * N$ rows. For example, Merkle tree partition with 4 levels and a fanout of 8, has 512 leaf nodes and 585 total nodes in the partition. We vary the number of partitions from 100 to 1000 in order to bring down the data conflicts between transactions by orders of magnitude.

9.2.3 Dynamic Merkle Tree. As discussed earlier in section 6, a fixed size merkle tree is not the most efficient to recover databases of different sizes. We now discuss how to scale the merkle tree dynamically with the user relation, keeping tuples per leaf node bounded within a range. The lower bound of the range is the node merge threshold whereas the node split threshold is the upper bound. It should not be too small (causes frequent merkle tree restructuring) nor too large (becomes more like static merkle tree). The internal data structure to maintain merkle tree is implemented in postgres as a relation.

Hash can be computed from tuple attributes, using any standard hashing function like Blake3 [35], MD5 [40], [50], SHA-512 etc. Tuple Hash used to map user relation tuples to leaf nodes, could either be the entire hash computed by the hashing function or a fixed subset of it, like a prefix or suffix or intermediate range, as long as same mechanism is used for entire merkle tree, across all replicas. Note that the hash length is chosen to be long enough so as to differentiate among tuples and it is not required to offer strong cryptographic security.

When splitting a large leaf node (as described in section 6) to create child leaf nodes, we can utilize additional bits from the tuple_hash, to distribute the mapping tuples to new child leaf nodes. Tuple hash could be implemented in Postgres either (a) as a new generated column in the user relation or (b) as a function index.

A binary merkle tree (fanout of 2) is many more levels deeper, compared to a merkle tree with larger fanout, which is shorter. When recovering a corrupt replica from a healthy replica, respective nodes of merkle tree are compared starting with root node and then all next level children of nodes with differing *node_hash*. For a

binary merkle tree, this will require many roundtrip messages, accessing merkle node information from memory and then sending it over the network. With other parameters constant, this latency is significantly reduced for merkle tree with larger fanout.

9.3 Corruption recovery

The system administrator inserts a special 'compareStates' transaction into the input log, to compare the database state at all replicas at the same instant in the transaction log, say P_{tx} . The transaction atomically (a) saves a logical database snapshot S_t when all preceding transactions have committed and (b) publishes as result, the root hash of the Merkle tree (or root hashes of all trees in the partitioned case). The snapshot S_1 includes all database relations, including the user relation and the corresponding Merkle trees.

As mentioned earlier, one of the healthy replicas is chosen as the reference replica N_{ref} . The recovery program simultaneously connects to replica N_{ref} with a read-only session and to the corrupt replica with a read-write session. On the corrupt replica, commands are sent to (a) disable new connections, (b) disconnect all connections other than the recovery session, and (c) disable / defer the constraints. On both replicas, (a) the database snapshot is set to S_t and (b) Merkle node hashes are requested (of all partitions in partitioned case).

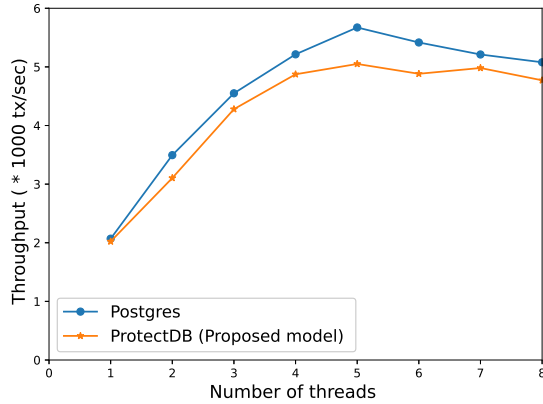
Merkle trees at the corrupted replica and the replica N_{ref} are recursively compared down from the root node, to find all leaf nodes whose hashes differ. Merkle trees are many levels deep and can be compared across replicas either (a) by fetching one level differing node hashes at a time or (b) by fetching entire tree in single request or (c) by fetching multiple levels at a time. Approach (a) results in too many round trips and thus increases data transfer time. Approach (b) is ideal but not feasible if entire merkle tree is not available in memory to transfer in single response. Approach (c) combines the advantages of the two approaches by balancing disk-fetch-time of merkle tree data with data-transfer time.

From mismatching leaf-nodes, we identify differing rows in the database by reverse-lookup using functional index on snapshot S_t . The differing rows are then replicated from snapshot S_t of replica N_{ref} . The state of the erstwhile corrupted replica is then brought to the current state by replaying the input transaction log starting from the point P_{tx} where the snapshot S_t was taken. This is very efficient except in case the corruption is major, when it is best to copy over a snapshot without Merkle tree comparison.

10 Performance Evaluation

We compare the performance of our proposed model on local and cloud setups, on the popular OLTP benchmark, YCSB. The local node in our setup is an AMD Ryzen7 5700G with 16 cores, NVMe SSD, and 16GB RAM running 64-bit Ubuntu 22.04 LTS. On AWS, results are obtained using *m5d.4xlarge* EC2 instances, with Ubuntu 24.04 and SSD storage with Linux kernel version 6.14.0-1018-aws and GCC version 13.3.0

Traffic generator Benchbase [12] generates transactions in a pre-configured proportion of operations. It supports multiple benchmarks like YCSB, Smallbank, and TPC-C, each of which defines its own schema.



Performance with signed queries, Merkle trees and passive detection: with and without determinism

Figure 9: Impact of Determinism

RSA key-pairs are generated with a common configuration (PKCSv15 padding, 2048 key-length, UTF-8 format, Base64 encoding, etc) for both client and system, and public keys are published to the destination .

10.1 Evaluation on single node

We first conduct an ablation study of different factors on a single machine.

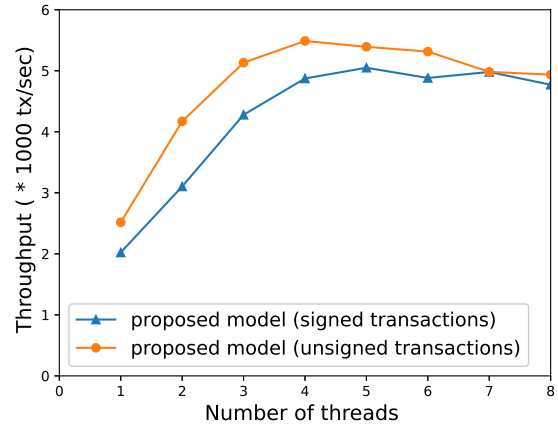
10.1.1 Impact of determinism. We compare our proposed deterministic model with the baseline of (non-deterministic) Postgres with Serializable transaction isolation, on a local machine with SSD storage.

As seen in Figure 9, the throughput initially increases linearly with number of threads. For higher number of threads (greater than 5), memory IO becomes a limiting factor, and contention between threads leads to a slight dip in throughput.

Our proposed deterministic database model gives a peak performance of around 5000 transactions per second, which is slightly lower than that of native PostgreSQL; performance is within 10% of baseline PostgreSQL across the number of concurrent threads considered. The performance closely matches that of other deterministic database implementations, as reported in Harmony[26].

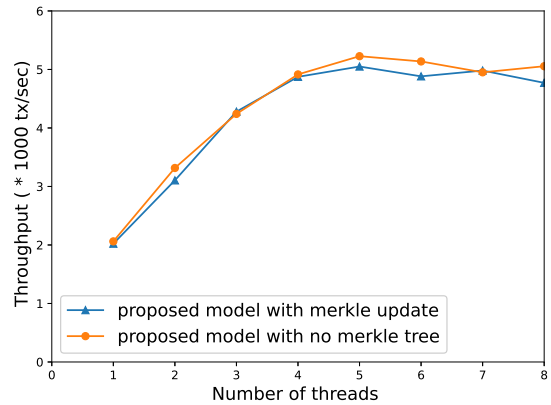
10.1.2 Impact of strict ordering . The performance of Aria[28] is evaluated as concurrency increases. Aria implementation provided by authors of [26], provides throughput of roughly 2200 transactions/sec (for block size of 1) and it improves to roughly 10000 transactions /sec (block size of 10) and stabilizes around 8000 transactions /sec (block size of 25). This is comparable to performance of our system, with the difference attributed to strict ordering.

10.1.3 Impact of signed transactions. As seen in Figure 10, adding signatures reduces throughput a little with a smaller number of threads, but as the number of threads increases, this difference vanishes. Since signature creation is CPU-bound it benefits from extra threads.



Performance with determinism, Merkle trees and passive detection: with and without signing of transactions

Figure 10: Impact of RSA-signing of transactions



Performance with determinism, passive detection, and signing of transactions: with and without Merkle trees.

Figure 11: Impact of maintaining Merkle tree

10.1.4 Impact of maintaining Merkle tree. For each user relation (e.g. usertable in YCSB), a Merkle tree is constructed with 200 partitions and 16 leaf nodes per partition. As seen in Figure 11, throughput increases linearly and then saturates in both cases, with and without Merkle tree. However, there is not much difference in system throughput in the two cases. This is probably because the Merkle tree nodes are frequently accessed and would be in buffer, so the IO overhead is minimal.

10.2 End-to-end system performance

We evaluate our proposed model on a multi-node setup. We measure the impact of adding Kafka alone, with transactions already serialized (as is the case in some applications), and the impact of adding Raft consensus for determining the serialization order along

Figure 12: Performance of End-to-end system

with Kafka. Our model needs Kafka to collect the execution results, so we do not consider the case of input-query-consensus alone.

Raft consensus algorithm, as implemented by Apache Ratis is used across 4 replicas. One of the replica nodes is configured with Kafka broker. A transaction is considered complete when its results are received by the Raft client from a threshold number of servers (in our case 3), and they match.

Figure 12 shows the throughput numbers for the single node case, compared with cluster with Kafka alone, and cluster with Raft and Kafka, with different numbers of threads at the client. We can see that on a cluster the performance with Kafka alone as well as with Raft and Kafka do not increase with number of threads, indicating that the system is not CPU bound. Both Kafka and Raft require multiple rounds of messaging, which appears to be the bottleneck. Further, using Raft reduces the throughput compared to using Kafka without consensus, indicating that Raft is currently the bottleneck.

10.3 Corruption Recovery performance

While actual corruption recovery would happen across replicas, we evaluate by hosting both healthy and corrupt databases on single node, since network I/O costs vary and are not relevant.

10.3.1 Other Corruption Recovery Approaches. We evaluate performance of proposed corruption recovery mechanism using YCSB benchmark with 1, 3 and 5 million tuples (244 MB, 750 MB and 1221 MB respectively). We compare the performance of our proposed merkle tree approach with (a) snapshotting technique described in [42], namely, create table as (CTA) and (b) Snapshot on disk: Save-to and restore-from disk file. Note that the compared approaches do not have provision to selectively restore the modified tuples, unlike our approach. Hence, the measurements for those approaches, are corresponding to complete backup and restore. The recovery time measurements of the static merkle tree approach correspond to the case where 30 merkle leaf nodes are corrupted. Since our approach detects corruption as soon as results from majority replicas are published, it is almost instantaneous. Thus recovery happens much sooner, before many nodes get corrupted. As shown in Figure 13, our approach performs recovery in nearly half the time of CTA approach, while the disk dump approach takes more than 4x longer.

10.3.2 Corruption Recovery optimizations. As described earlier, in our proposed merkle tree based approach to corruption Recovery majority of time is spent in (a) fetching the merkle trees from disk and (b) comparing across replicas across network and (c) fetching the rows which map to differing merkle leaf nodes, from disk and finally (d) updating the corrupt data on disk.

We optimize the transactions to recover the compromised replica by observing query plans, batching and then apply to restore corrupt data. To reduce the overhead in fetching both merkle and user data from both replicas, we use AsyncConnectionPool package to concurrently execute the queries to both replicas, asynchronously. Figure 15 shows the recovery time of database using our approach, with and without optimizations.

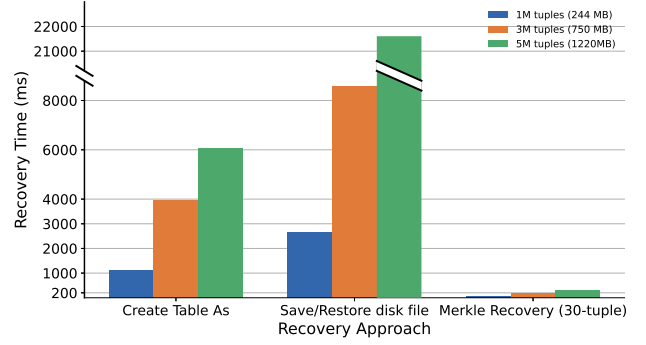


Figure 13: Performance comparison of Corruption Recovery approaches

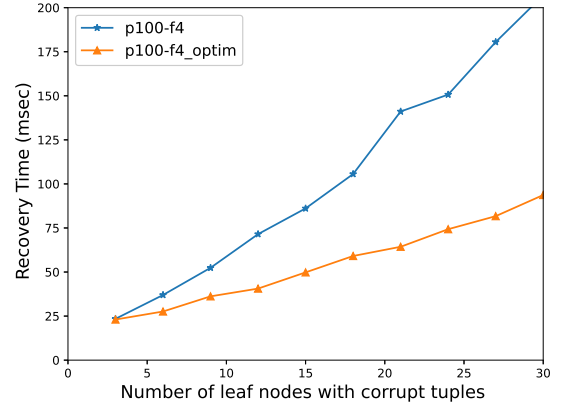


Figure 14: Performance of Database Corruption Recovery with and without optimizations

The comparison of merkle trees described earlier, compares merkle nodes across replicas at one-level, finds differing nodes and then repeats for their next-level child nodes. This can be optimized by comparing all nodes which are higher-order descendants (multiple level down) of differing nodes. This is especially helpful when the merkle tree fanout is small (e.g. 4 or 8) or network latency between replicas is high, and comparable to disk read latency. However, for co-located replicas, with large merkle tree fanout (say 32), fetching 2-level data (1024 nodes) take much longer (800ms) compared to two one-level lookups of differing nodes (150ms each). For taller merkle trees with small fanout, the lookup overheads make multi-level lookups more efficient.

10.3.3 Corruption Recovery Ablation Study . We evaluate the impact of different factors like number of merkle tree partitions, merkle tree fanout, number of keys per leaf node.

Note that if at least one corrupted tuple maps to each Merkle tree leaf node, all rows mapping to that leaf node need to be compared to determine if it is corrupted and restored. To simulate corruption of k leaf nodes, corrupted tuples are distributed uniformly among the k leaf nodes, preferably belonging to different partitions. This finds the worst performance of system by requiring exploration

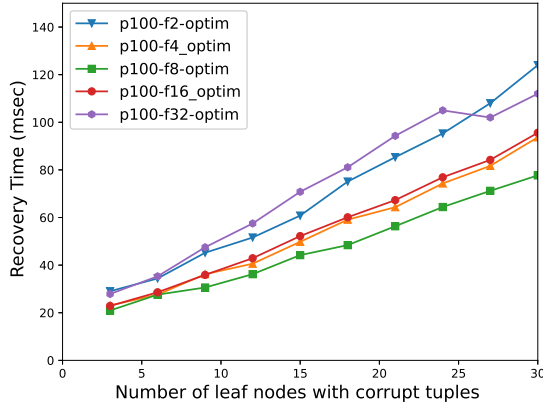


Figure 15: Performance of Database Corruption Recovery for different fanout

of path to each leaf node from its partition-root. The user relation tuples are corrupted through insert, delete and update operations which are equally likely.

For example, with fanout of 32 and 2-levels, we have 1024 leaf nodes per partition and with 100 Merkle partitions, nearly 30 tuples map to a Merkle leaf node for user relation of 3M tuples.

Figure 15 shows the recovery time of database using our approach with varying fanout. Observe that the recovery time first decreases as fanout increases from 2 to 8 and then increases with fanout.

Figure 16 shows the recovery time of database using our approach with varying number of partitions, which shows a similar pattern as observed with varying fanout. Recovery time first decreases from 50 to 100 partitions but again increases for 200 partitions.

Figure 17 shows the recovery time of database using our approach where all corrupted tuples are (a) in same merkle leaf node (b) in same merkle tree partition and (c) in distinct partitions. As expected, time to recover single leaf is almost constant. As more leaf nodes in same partition are corrupted, the recovery time grows. When leaf nodes are in distinct partitions, recovery take longer than other 2 cases.

The recovery time increases with number of leaf nodes having corrupted tuples. This is due to proportionately more number of tuples fetched and compared. Also, note that with fewer partitions of merkle tree, more tuples map to a leaf node. Thus for same number of leaf nodes corrupted, more tuples (mapping to corrupted leaf nodes) need to be fetched and compared, resulting in higher recovery time. The optimal parameters of merkle tree are expected to change, depending on the number of tuples in relation.

11 Future work

In many applications, most of the transactions update a small amount of data. Hence, it is possible to scale the system further, by sharding the dataset so that reads and updates of most transactions can be satisfied at a single shard. Each shard may be hosted at a subset of cluster replicas instead of all. For transactions which will access or update multiple shards,

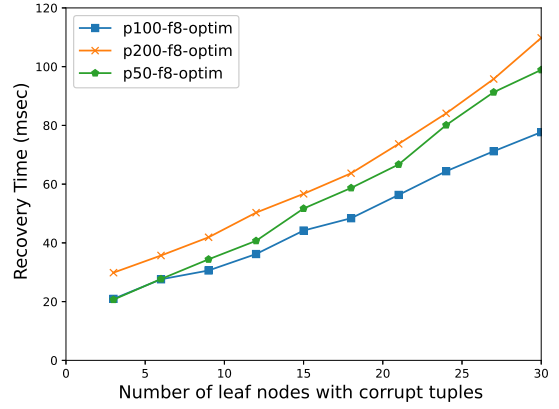


Figure 16: Performance of Database Corruption Recovery for different partitions

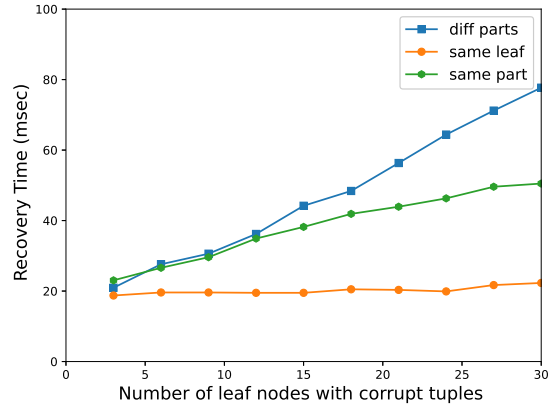


Figure 17: Performance of Database Corruption Recovery for spread of corruption

Partitioning the database enables to scale the system further. In Calvin [49] model, remote partition read stage is completed prior to executing transaction write stage. This model is applicable for our design as well.

Another possible model is where all reads are executed first and then all writes are committed. This limits the concurrency as the reads of next transaction in sequence depend on writes of the current transaction.

12 Conclusion

We presented a design for protecting database data from corruption, based on the Byzantine-fault tolerant replicated state machine approach. As long as only a minority of replicas are compromised, our design supports recovery of the corrupt replicas while the system is fully operational, without any downtime. We show that the approach can be implemented quite efficiently, on top of PostgreSQL, with relatively small changes to PostgreSQL itself.

We observe that signature verification, Merkle tree maintenance and support for determinism impose relatively small overheads on

transaction processing throughput. On the other hand, both distributed consensus and Kafka add significant overheads on system throughput. Optimizing these components to improve throughput is an area of ongoing work. The system provides throughput of over 5000 YCSB transactions per second, and recovery is shown to be very efficient.

We believe that our approach will have significant practical impact, given the increased risks of attacks such as ransomware or advanced penetration attacks that compromise servers.

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